

Discrete Time Markov Chains (DTMC)

= a stochastic process $\{X_n, n \in \mathbb{Z}^+\}$ such that for any n ,
 $P(X_n = j \mid X_1 = x_1, \dots, X_{n-1} = x_{n-1}) = P(X_n = j \mid X_{n-1} = x_{n-1})$

= MARKOV PROPERTY

$$P(\text{next state} \mid \text{past states, present state}) = P(\text{next state} \mid \text{present state})$$

Example: Coin with memory (Markovian coin)

$$S = \{+1, -1\}$$

$\nearrow$ heads
 $\searrow$ tails

Sticky coin: $P(X_{n+1} = 1 \mid X_n = 1) = 0.9$
 $P(X_{n+1} = -1 \mid X_n = -1) = 0.8 \quad \forall n$

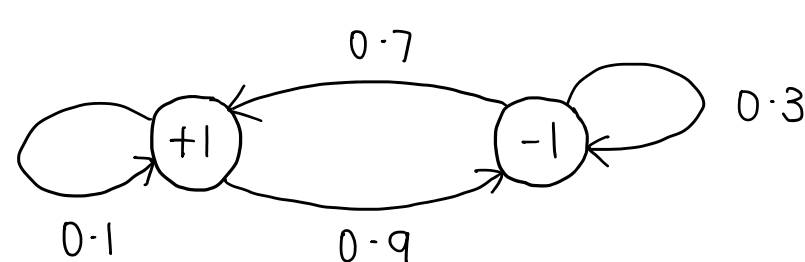
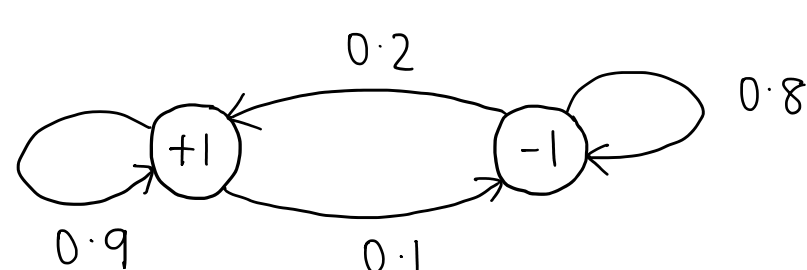
Flippy coin: $P(X_{n+1} = 1 \mid X_n = 1) = 0.1$
 $P(X_{n+1} = -1 \mid X_n = -1) = 0.3 \quad \forall n$

Transition probability matrix P

$$X_n \begin{matrix} X_{n+1} \\ \begin{bmatrix} 0.9 & 0.1 \\ 0.8 & 0.2 \end{bmatrix} \end{matrix}$$

$$X_n \begin{matrix} X_{n+1} \\ \begin{bmatrix} 0.1 & 0.9 \\ 0.3 & 0.7 \end{bmatrix} \end{matrix}$$

Transition diagram



Finite Dimensional Distributions

Consider a Markov dice with transition probability P

What is $P(X_0 = 4, X_1 = 5, X_2 = 6)$?

$$= P(X_2 = 6 \mid X_1 = 5, X_0 = 4) \cdot P(X_1 = 5 \mid X_0 = 4) \cdot P(X_0 = 4)$$

$$= p_{56} p_{45} \cdot P(X_0 = 4)$$

What is $P(X_0 = 4)$?

The probability of starting in a particular state is called the initial distribution of the Markov chain.

Assume M states.

Let $\bar{\mu} = (\mu_1, \dots, \mu_M)$ denote the initial distribution.
 i.e. $P(X_0 = x_0) = \mu_{x_0}$

$$P(X_0 = x_0, X_1 = x_1, X_2 = x_2) = p_{x_1, x_2} \cdot p_{x_0, x_1} \cdot \mu_{x_0}$$

In general, $P(X_0 = x_0, X_1 = x_1, \dots, X_k = x_k) = p_{x_{k-1}, x_k} \times \dots \times p_{x_0, x_1} \times \mu_{x_0}$

Chapman Kolmogorov Equations for DTMC

Consider a Markov coin and its transition probability matrix

$$P = \begin{bmatrix} p_{1,1} & p_{1,-1} \\ p_{-1,1} & p_{-1,-1} \end{bmatrix}$$

$$P(X_2 = 1 \mid X_0 = 1) = P(X_2 = 1 \mid X_1 = 1) \cdot P(X_1 = 1 \mid X_0 = 1) + P(X_2 = 1 \mid X_1 = -1) \cdot P(X_1 = -1 \mid X_0 = 1)$$

$$= p_{1,1} \times p_{1,1} + p_{-1,1} \times p_{1,-1}$$

$$= p_{1,1}^2 + p_{-1,1} \cdot p_{1,-1}$$

$$P(A) = P(A|B) \cdot P(B) + P(A|B^c) \cdot P(B^c)$$

$$P(A|C) = P(A|B \cap C) \cdot P(B|C) + P(A|B^c \cap C) \cdot P(B^c|C)$$

Similarly, $P(X_2 = -1 \mid X_0 = 1)$, $P(X_2 = 1 \mid X_0 = -1)$ and $P(X_2 = -1 \mid X_0 = -1)$ can be obtained.

These are elements of a two-step transition matrix $P^{(2)}$.

$$P^{(2)} = P \times P = P^2$$

In general, $P^{(n)} = P^n$

Chapman-Kolmogorov equations are a further generalisation of this.

$$p^{(n+l)} = p^{(n)} \cdot p^{(l)}$$

$$X_n \begin{matrix} X_{n+1} \\ \begin{bmatrix} p \end{bmatrix} \end{matrix} \quad X_n \begin{matrix} X_{n+5} \\ \begin{bmatrix} p^4 \end{bmatrix} \end{matrix}$$